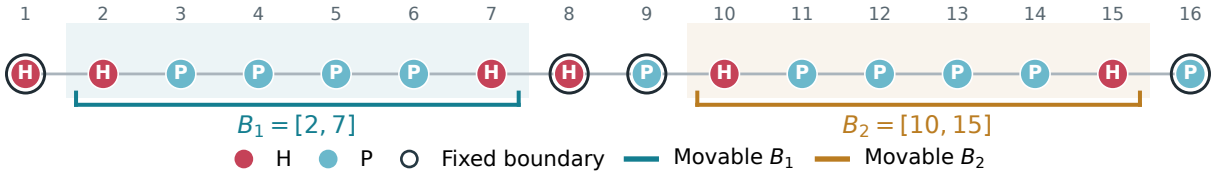
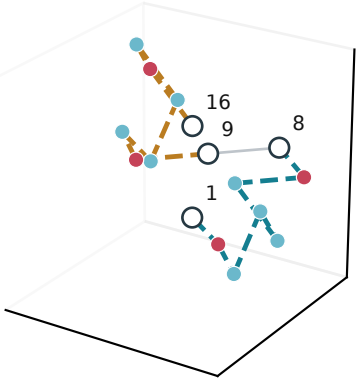
(a) Illustrative HP sequence and fixed boundaries

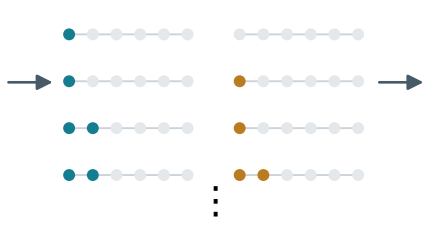


(b) Current fold: $C_{HH} = 0$



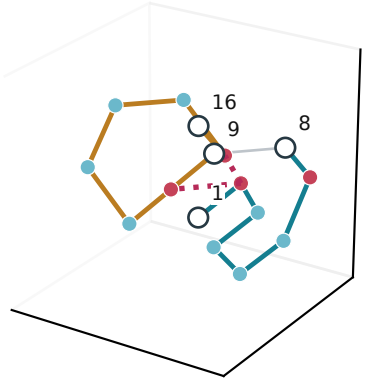
(c) Joint reconstruction

One partial state (schematic)



Alternate extensions; enforce both endpoint links and no overlap

(d) Shown draw: $C_{HH} = 6$



Gray chain stays fixed; black rings identify the same four anchors in both folds.

(e) Direct heat-bath draw from all 25 feasible pool members ($T = 1.2$, $\lambda_s = 0.66$)

Pool entry	C_{HH}	C_{ex}	F_s	p_j
Current fold	0	0	0.00	<0.0001
Shown draw	**6**	**2**	**-12.66**	**0.2008**
Another alternative	6	2	-12.66	0.2008
Other 22 alternatives	--	--	--	0.5983